# Differentiation Rules: Power, Product, Quotient, Chain

ANSWER KEY

## Power, sum, and constant rules

**1** Differentiate each function:

**a** $f(x) = 5x^4 - 3x^2 + 7$
$= 20x^3 - 6x$

**b** $f(x) = \frac{1}{x^3} \quad = -3x^{-4}$

**c** $f(x) = 4\sqrt{x} \quad = \frac{2}{\sqrt{x}}$

## Product and quotient rules

**2** Differentiate $f(x) = (2x+1)(x^2-3)$ using the product rule.
$f'(x) = 2(x^2-3) + (2x+1)(2x) = 2x^2 - 6 + 4x^2 + 2x = 6x^2 + 2x - 6$.

**3** Differentiate $f(x) = \frac{x^2+1}{x-2}$ using the quotient rule.
$f'(x) = \frac{2x(x-2) - (x^2+1)(1)}{(x-2)^2} = \frac{x^2 - 4x - 1}{(x-2)^2}$.

**4** Differentiate $f(x) = x^2 \sin x$.
$f'(x) = 2x\sin x + x^2\cos x$.

## The chain rule

**5** Differentiate each function using the chain rule:

**a** $f(x) = (3x^2-5)^4$
$= 4(3x^2-5)^3(6x) = 24x(3x^2-5)^3$

**b** $f(x) = \sin(2x^3) \quad = 6x^2\cos(2x^3)$

**c** $f(x) = e^{-x^2} \quad = -2xe^{-x^2}$

**d** $f(x) = \ln(x^2+1) \quad = \frac{2x}{x^2+1}$

**6** Differentiate $f(x) = \sqrt{4x + \cos x}$.
$f'(x) = \frac{4 - \sin x}{2\sqrt{4x + \cos x}}$.

## Combined rules

**7** Differentiate $f(x) = \frac{\sin x}{x^2}$ using the quotient rule.
$f'(x) = \frac{x^2\cos x - \sin x(2x)}{x^4} = \frac{x\cos x - 2\sin x}{x^3}$.

**8** Differentiate $f(x) = e^{2x}\cos(3x)$ using the product and chain rules together.
$f'(x) = 2e^{2x}\cos(3x) + e^{2x}(-3\sin 3x) = e^{2x}(2\cos 3x - 3\sin 3x)$.

**9** Differentiate $f(x) = \left(\frac{x+1}{x-1}\right)^3$ using the chain rule together with the quotient rule.
Let $u = \frac{x+1}{x-1}$; $\frac{du}{dx} = \frac{(x-1)-(x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2}$. So $f'(x) = 3u^2 \cdot \frac{du}{dx} = 3\left(\frac{x+1}{x-1}\right)^2 \cdot \frac{-2}{(x-1)^2} = \frac{-6(x+1)^2}{(x-1)^4}$.

## Tangent lines using derivatives

**10** Find the equation of the tangent line to $f(x) = x^2 - 3x + 1$ at $x = 2$.

$f(2) = 4 - 6 + 1 = -1$; $f'(x) = 2x - 3$, $f'(2) = 1$. Tangent: $y - (-1) = 1(x - 2)$, i.e. $y = x - 3$.

**11** Find the point(s) on the curve $y = x^3 - 3x$ where the tangent line is horizontal.

$y' = 3x^2 - 3 = 0$ gives $x = \pm 1$. Points: $(1, -2)$ and $(-1, 2)$.

## Higher-order derivatives

**12** Find $f''(x)$ for $f(x) = x^4 - 6x^2 + 2x$.

$f'(x) = 4x^3 - 12x + 2$; $f''(x) = 12x^2 - 12$.

**13** Find $f''(x)$ for $f(x) = \sin(2x)$.

$f'(x) = 2\cos(2x)$; $f''(x) = -4\sin(2x)$.

## Normal lines

**14** Find the equation of the line normal (perpendicular) to $f(x) = x^2$ at $x = 2$.

$f(2) = 4$; $f'(x) = 2x$, $f'(2) = 4$, so the tangent slope is $4$ and the normal slope is $-\frac{1}{4}$. Normal line: $y - 4 = -\frac{1}{4}(x - 2)$, i.e. $y = -\frac{1}{4}x + \frac{9}{2}$.